✍️ Smart City Architecture (Layer-wise Description)

🔹 1. Street Layer (Bottom Layer)

Components:

- Smart lights
- Multi-sensing nodes
- Parking sensors
- Traffic flow sensors
- Cameras (FR – Face Recognition, LPR – License Plate Recognition, VA – Video Analytics)
- Mobile devices

Function:

- This layer **collects real-time data** from the environment
- Sensors and devices monitor traffic, parking, lighting, and security

👉 **Write in exam:**
"Street layer consists of sensors and devices that collect real-time data such as traffic, parking, and surveillance information."

🔹 2. City Layer

Components:

- Distribution switches

Function:

- Connects all street-level devices
- Transfers data to upper layers

👉 **Write in exam:**
"City layer acts as an intermediate layer that connects field devices and forwards data through distribution switches."

🔹 3. Data Center Layer

Components:

- Internet Gateway / Router
- ASA Firewall
- Wireless Controller
- DC Switch
- Compute & Storage Servers
- Virtualization
- Network Management
- Location Services Engine

Function:

- Processes and stores data
- Provides security (firewall)
- Manages network and services

👉 **Write in exam:**
"Data center layer processes, stores, and secures data using servers, firewall, and network management systems."

🔹 4. Service Layer (Top Layer)

Components:

- Parking Management
- Lighting Management
- Safety & Security
- Traffic Management
- Applications & Dashboards

Function:

- Provides services to users
- Displays data in dashboards
- Helps in decision-making

👉 **Write in exam:**
"Service layer provides smart city services like traffic control, parking management, and security through applications and dashboards."

🔚 Conclusion (Write this line for extra marks):

“This layered architecture enables efficient data collection, processing, and service delivery, making cities smarter and more efficient.”
